

Interference Filter External Cavity Diode Laser (ECDL) with a Cateye Configuration

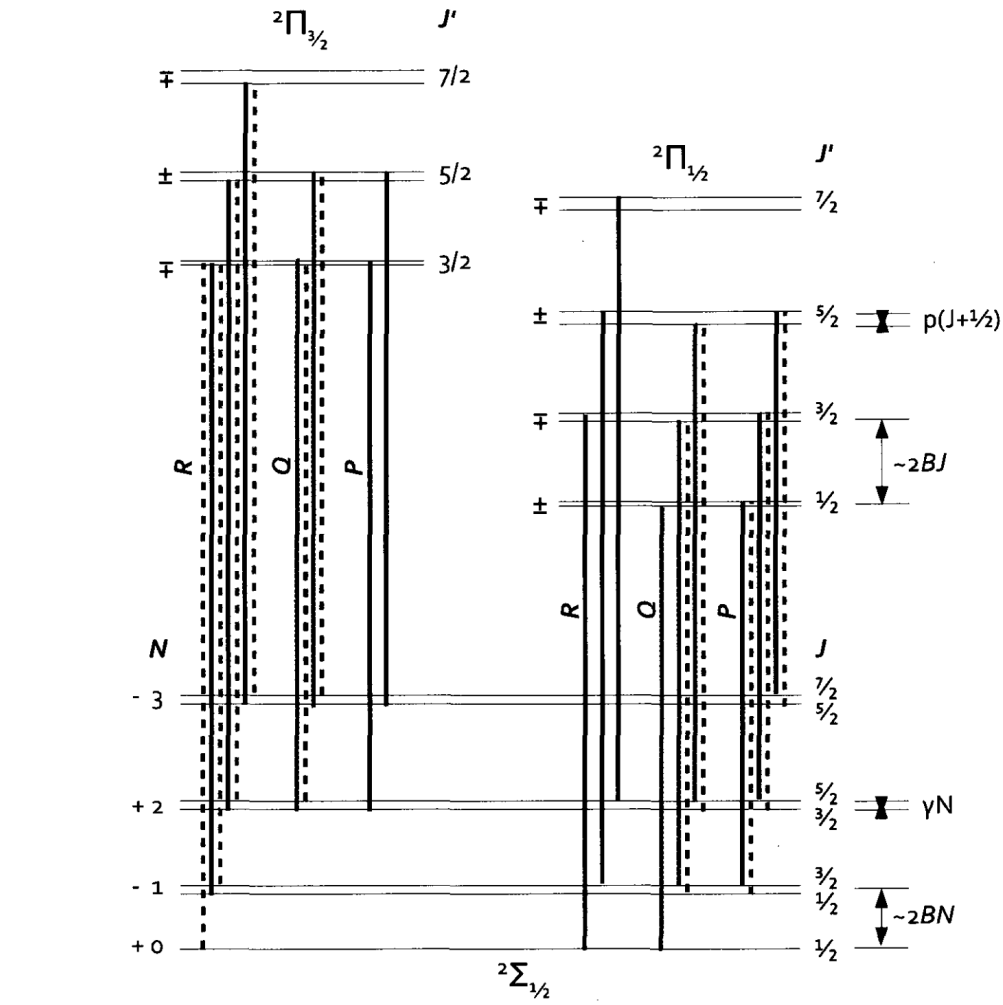
Robert Xi¹, Geoffrey Zheng¹, Mangesh Bhattarai², David DeMille^{1, 2}
1. James Franck Institute and Department of Physics, University of Chicago, IL 60637
2. Physics Division, Argonne National Laboratory, Argonne IL 604039

Goals and Motivation

ZOMBIES: Use ¹³⁷Ba¹⁹F to measure nuclear spin-dependent parity violation (NSD-PV) effect

Rotational Cooling (Pump population into single hyperfine level)

- Laser optically pumps ¹³⁷Ba¹⁹F molecules from $X^2\Sigma(v=0; N^P=0^+, 1^-, 2^+)$ to $A^2\Pi_{1/2}(J^P=5/2^+)$, leaving $X^2\Sigma(v=0; N^P=1^-, J=3/2)$ hyperfine level unaddressed.
- Spontaneously decay back to ground state, accumulate population to $X^2\Sigma(v=0; N^P=1^-, J=3/2)$.
- Simulation shows **×30** gain in the $X^2\Sigma(v=0; N^P=1^-, J=3/2)$ population.



Laser Requirements

- Narrow linewidth (~ 100 kHz)
Similar Level of precision would mean measuring earth-moon distance with an uncertainty of 1mm!
- Wide tunability (~ 30 GHz)
- High stability.
- Power output (~ 100 mW)
- Ease of use, small size, reproducible.

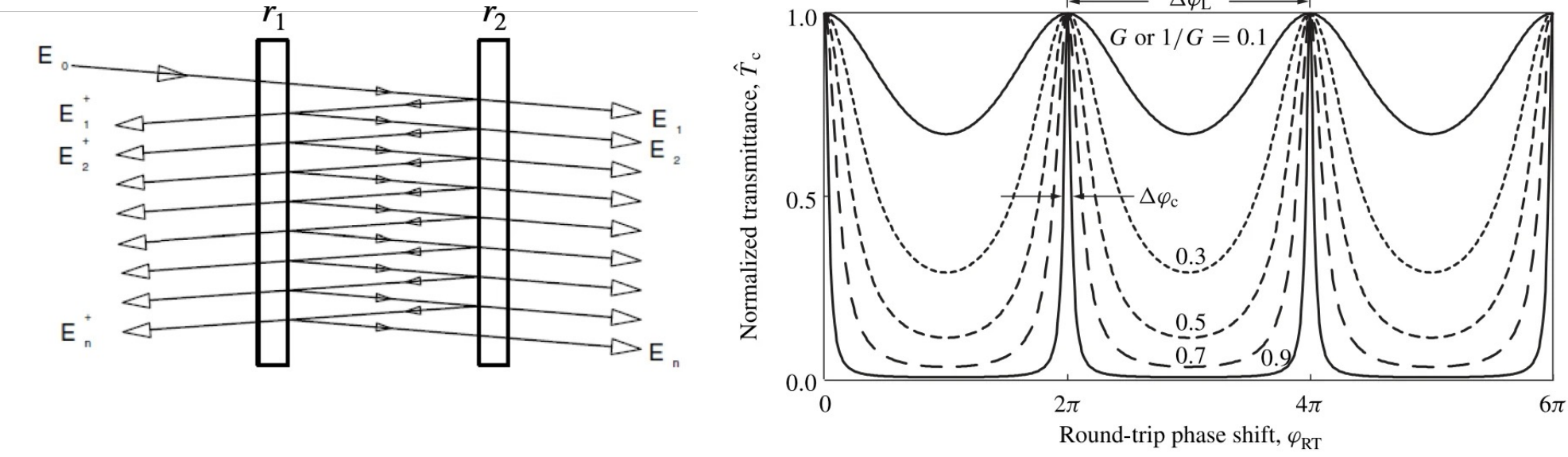
Theoretical Basis

Fabry-Perot Cavity

An optical resonator consisting of **two parallel, highly reflective mirrors** that reflect light back and forth, forming standing waves at specific resonant frequencies.

$$T_{\text{cavity}}(\nu, L) = \frac{1}{1 + F \sin^2 \delta(\nu)}$$

where F is the coefficient of **finesse** for the cavity. $\delta(\nu) = 2\pi nLv/c$, L is the **cavity length**, and n is the refractive index of the cavity medium.

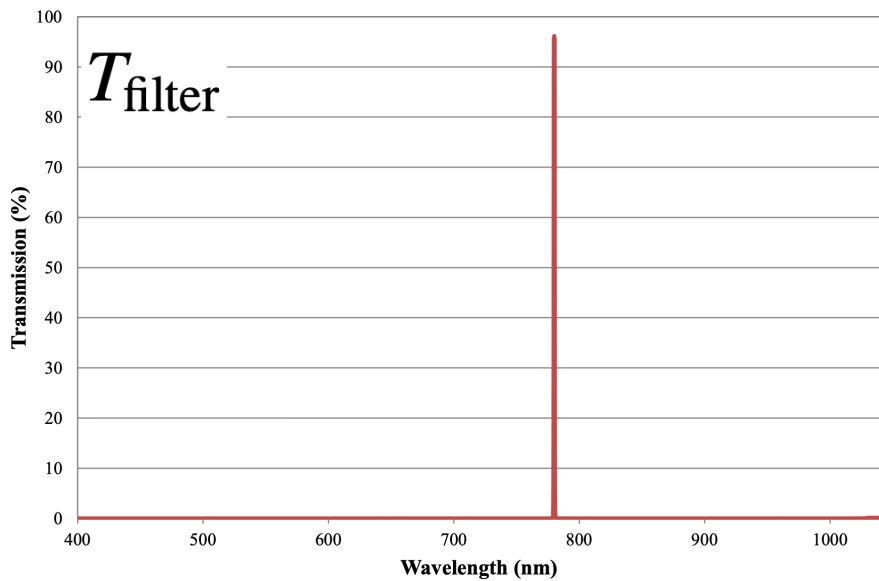


Ultra-narrow Bandpass Interference Filter

An optical filter that transmits light at specific wavelength (~ 1 nm) using interference from multiple thin layers.

The transmission window wavelength shift with incident angle θ :

$$\lambda(\theta) = \lambda_0 \sqrt{1 - \left(\frac{\sin(\theta)}{n} \right)^2}$$



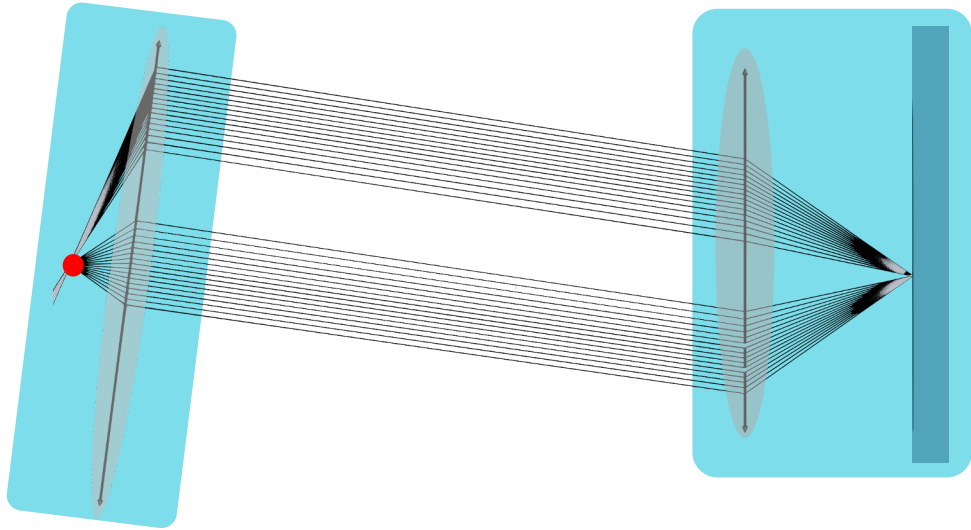
Cateye Retroreflector

A **lens-mirror combination (Cateye)** and a **lens-source** aligned facing each other, with the mirror/light source at the focal plane/point of the lens.

Reflects any incoming collimated beam back to the source, regardless of their relative angle (provided the beam strikes the Cateye).

Advantages:

- Easy to align
- Mechanically stable

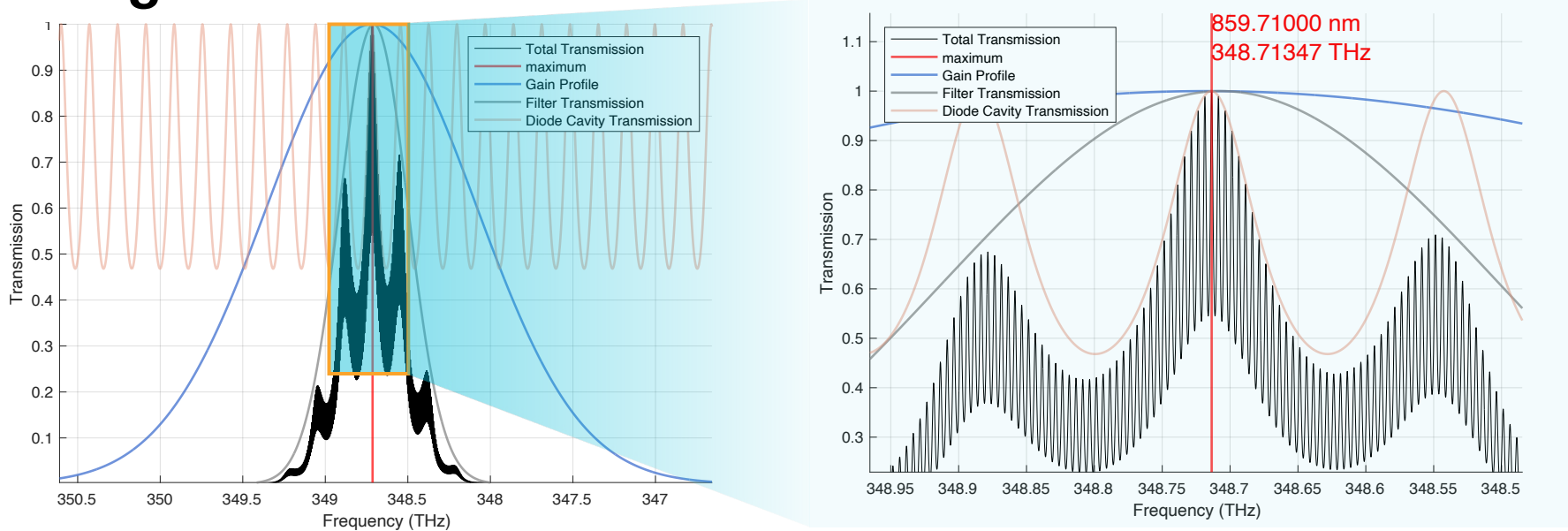


Laser Output Frequency

The output frequency ν of an ECDL depends on a combination of frequency-dependent gain and loss factors. The product of these factors gives the total transmission function:

$$T_{\text{total}} = G_D T_D T_{\text{cavity}} T_{\text{filter}}$$

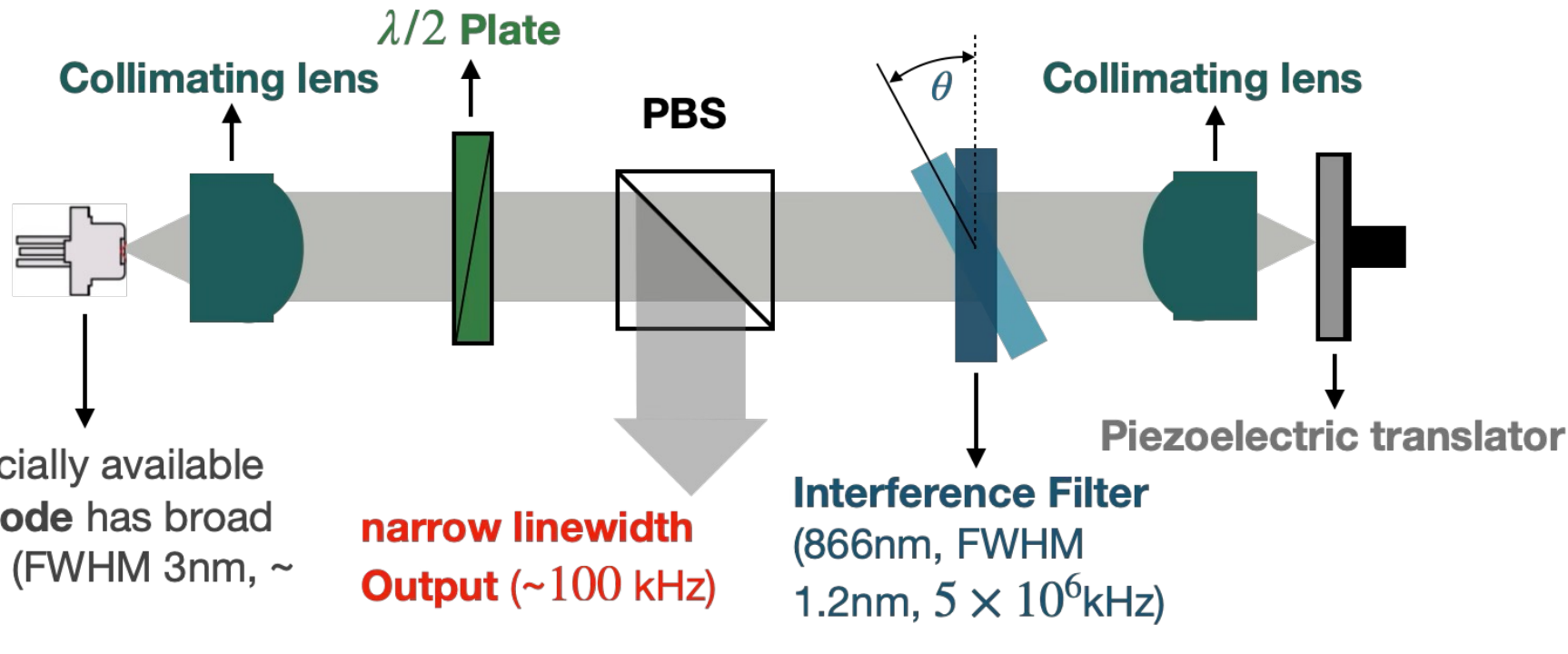
The laser will oscillate at the frequency for which the product of these factors is **greatest**.



Design

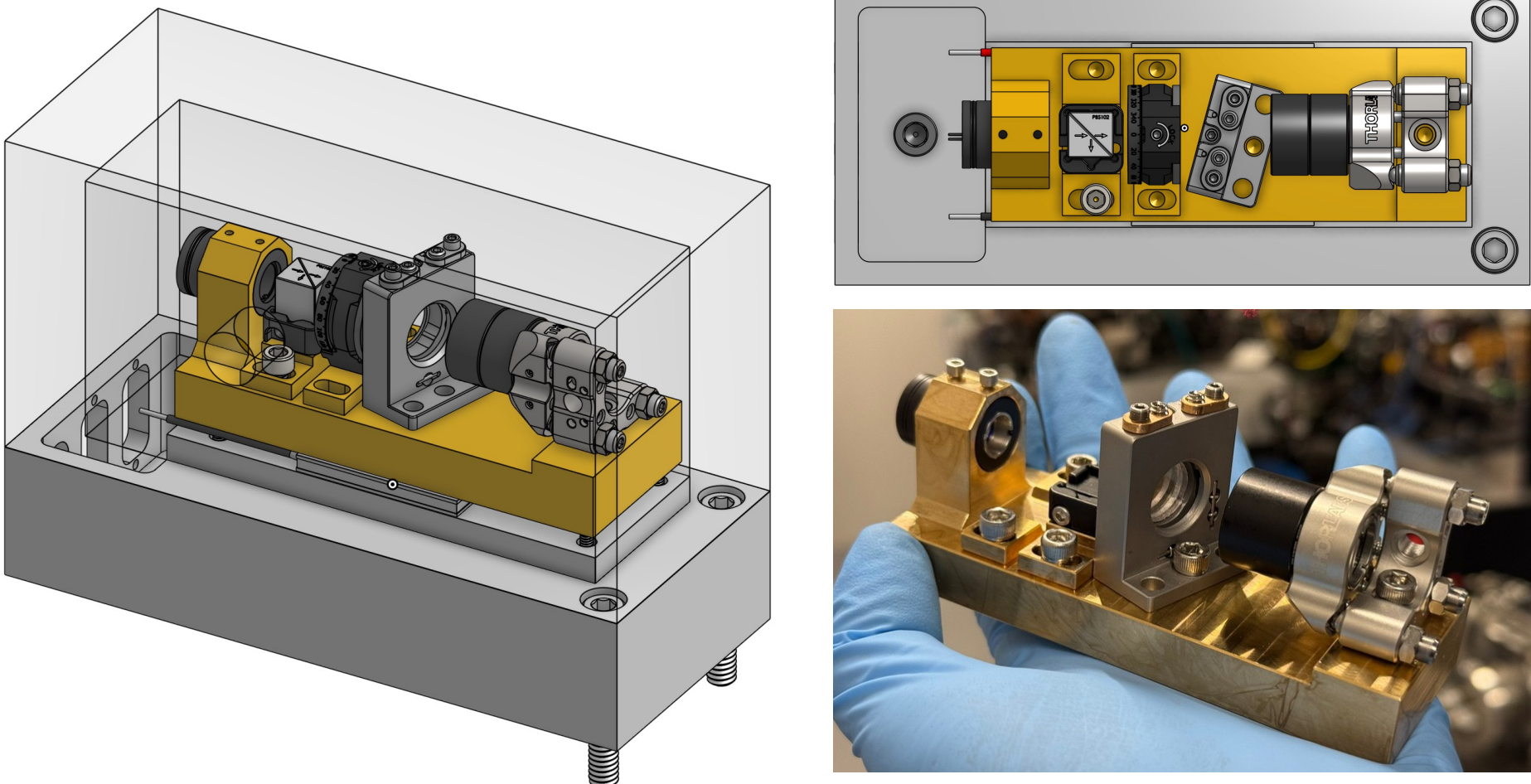
Design Principles

- $\lambda/2$ waveplate + PBS allows tuning the feedback intensity
- Self-aligning Cateye reflection geometry, mechanically robust
- Frequency coarse tuning by changing diode temperature (100GHz)
- Moderate tuning by changing laser diode current (10GHz)
- Fine tuning by changing piezo voltage (<1 GHz)



CAD

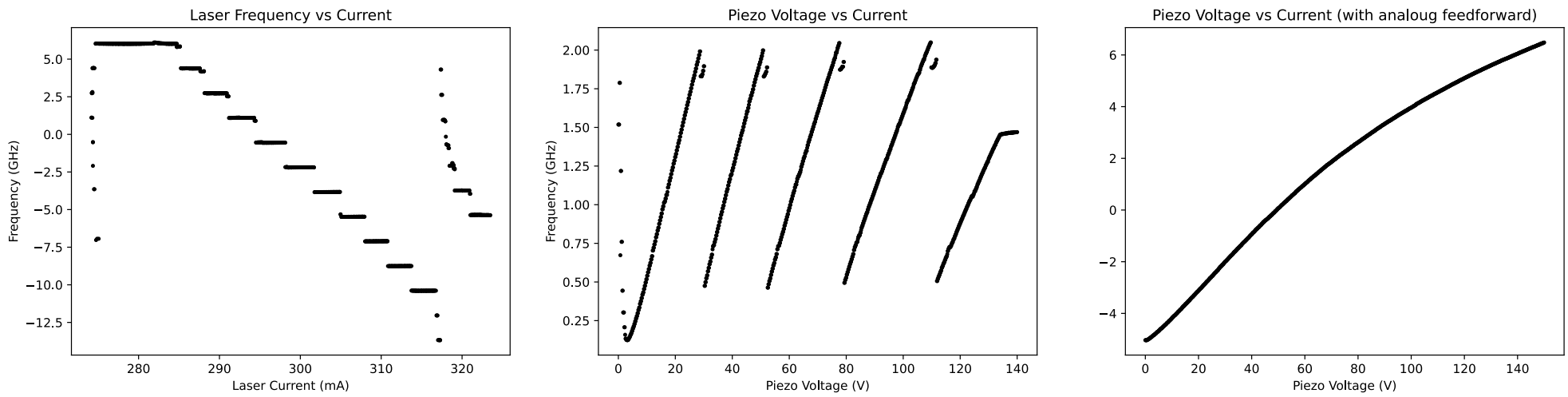
- Precise pitch and roll control of Cateye configuration allows finer tuning of alignment
- 1-inch-thick acrylic lid ensures system's thermal stabilization
- Slotted PBS and $\lambda/2$ waveplate brass spacer allows adjustment of their position
- Brass base provides good thermal conductivity



Result

Diode Current Scan, Piezo Voltage Scan, and Feedforward

- Without feedforward (scanning current and voltage simultaneously at certain ratio), scanning current gives discrete steps, scanning voltage gives ~ 1.5 GHz MHFTR, consistent with simulation
- With feedforward, MHFTR ~ 12 GHz



Outlook

Transfer Cavity for frequency stabilization and frequency shifting laser

